Project Synopsis

on

# A tool for lecture assessment as per OBE policy

Submitted as a part of course curriculum for

## Bachelor of Technology

in

## Computer Science

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# ABSTRACT

In contemporary educational settings, the integration of technology has become essential for fostering student engagement and enhancing learning outcomes. This project proposes the development of a dedicated hardware device designed to assist teachers in facilitating Multiple Choice Question (MCQ) assessments, thereby promoting active participation among students while mitigating distractions posed by mobile devices.

The device will provide an interactive platform for students to respond to MCQs in real time, allowing educators to gauge understanding and adjust their teaching strategies accordingly. By eliminating the need for smartphones or tablets during assessments, the device aims to create a focused learning environment, reducing the prevalence of off-task behaviour.

Key features of the device will include a user-friendly interface, instant feedback capabilities, and comprehensive analytics to track student performance over time. Teachers will be able to create, modify, and deploy question sets tailored to their curriculum, enhancing the relevance of assessments. Additionally, the device will support a variety of question types and provide instantaneous results, enabling educators to address learning gaps swiftly.

This project aligns with the principles of Outcome-Based Education (OBE), where the emphasis is placed on achieving specific learning outcomes. By equipping teachers with a reliable tool to engage students in MCQ assessments, the project seeks to foster a more interactive and effective learning experience, ultimately contributing to improved educational outcomes and student accountability. The success of this initiative will not only empower educators but also enhance student engagement, paving the way for a more productive and focused classroom environment.

# INTRODUCTION

In the contemporary educational landscape, mobile devices have become ubiquitous, often serving as double-edged swords. While they offer access to a wealth of information and tools, they are also major sources of distraction, particularly in classrooms. Teachers face the challenge of maintaining student engagement and focus, especially during activities like quizzes or assessments where mobile phones are frequently used. This reliance on mobile devices can lead to decreased attention, disruptions in learning, and difficulties in accurately assessing students' comprehension.

To address this pressing issue, our project introduces a specialized hardware device that offers a distraction-free solution, designed specifically for Outcome-Based Education (OBE). The device is intended to be used by students during lessons, allowing them to answer Multiple Choice Questions (MCQs) in a controlled and focused environment. By removing the need for mobile phones, it helps ensure that students remain attentive and actively participate in classroom activities, thus creating a more productive and engaging learning atmosphere.

What sets this device apart is its seamless integration with an accompanying software system. This software will evaluate and track each student's responses in real-time, providing instant feedback to teachers. The data generated will offer insights into individual student performance, allowing educators to identify learning gaps and tailor instruction to meet the needs of their students. Additionally, this system aligns with the principles of OBE, which emphasizes measurable outcomes and continuous improvement in teaching and learning.

By merging hardware and software into a comprehensive solution, this project not only addresses the issue of classroom distractions but also enhances the effectiveness of learning assessments. Ultimately, it aims to support teachers in creating a focused, interactive, and outcome-driven learning environment.

# PROBLEM STATEMENT

Smart Education refers to the integration of digital tools in learning, aimed at making education more effective, efficient, and flexible. In this approach, students have access to a wide range of resources that cater to their individual learning needs, enabling them to study comfortably and at their own pace. However, the reliance on mobile devices in classrooms has also introduced new challenges.

While mobile devices provide access to interactive learning materials, they are also significant sources of distraction. Notifications, social media, and non-educational apps can easily divert students' attention away from lessons, reducing focus and overall classroom engagement. This issue becomes particularly prominent during assessments, such as Multiple-Choice Question (MCQ) quizzes, where students’ attention is crucial for accurate performance evaluation. The constant distractions from mobile devices can compromise the integrity of assessments and reduce the quality of learning outcomes.

To fully harness the potential of Smart Education, it is essential to develop tools that minimize these distractions while maintaining the advantages of digital learning. A dedicated hardware device, separate from mobile phones, could provide a focused learning environment for students, enabling them to participate in assessments and quizzes without external interruptions. This solution will not only enhance classroom engagement but also ensure more accurate and reliable assessment of student performance.

# OBJECTIVES

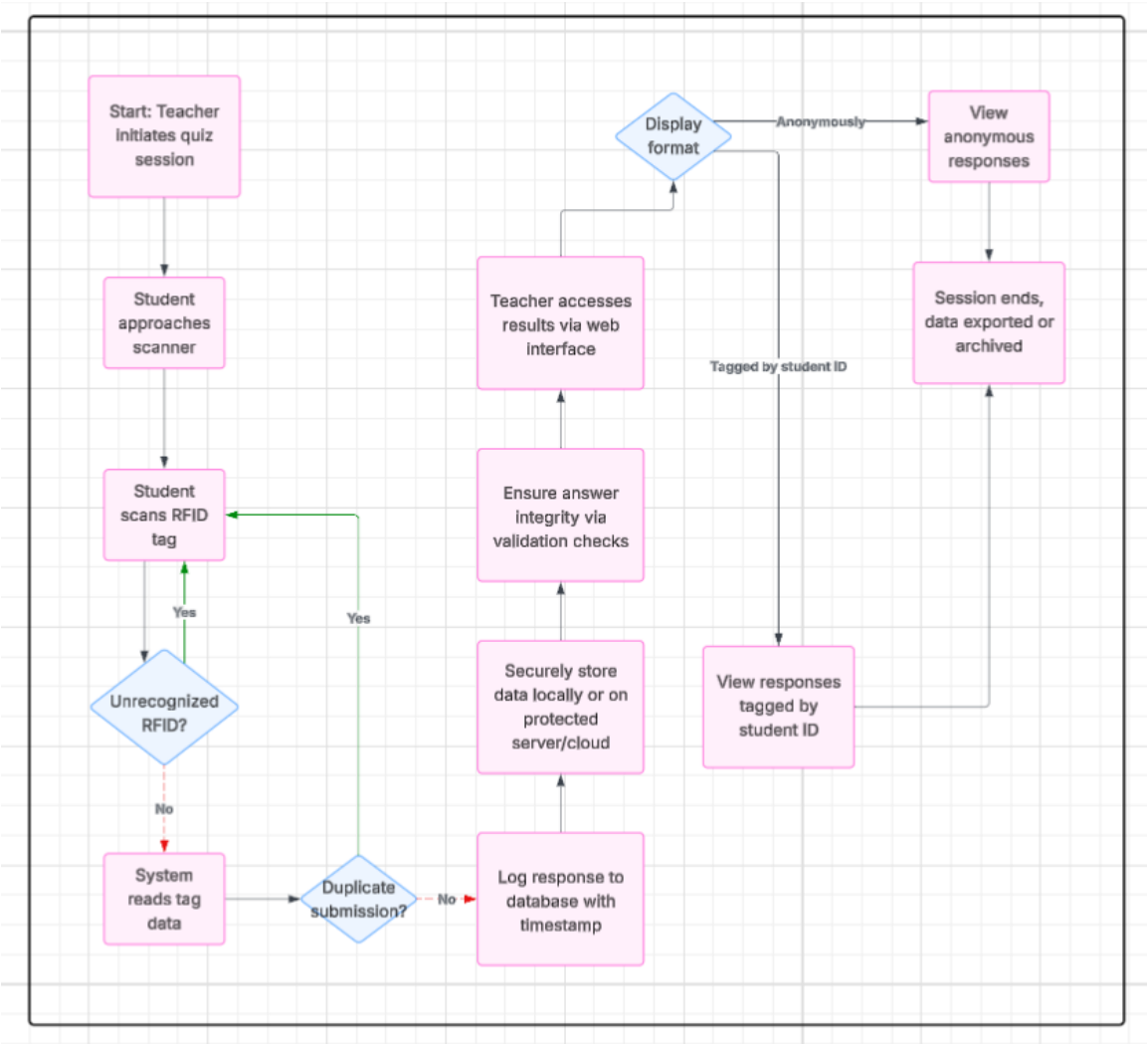
The objective of this project is to design and develop a specialized hardware device, integrated with an accompanying software system, to facilitate Multiple Choice Question (MCQ)-based assessments in classrooms. This device is specifically designed to minimize distractions by eliminating the need for mobile phones, which often lead to reduced student focus and engagement.

In line with the principles of Outcome-Based Education (OBE), the goal is to create a tool that allows for real-time, interactive participation during assessments, helping educators evaluate student understanding more accurately. The hardware will provide students with a focused platform for answering MCQs during lessons, ensuring that they are not distracted by notifications, social media, or other non-educational activities.

The integrated software will track and evaluate each student's responses, offering instant feedback to teachers. This real-time data will help educators identify learning gaps, measure student performance against learning outcomes, and adapt their teaching strategies accordingly. By fostering active participation and delivering real-time assessments, the project aims to create a more engaged and outcome-driven learning environment.
Ultimately, the project seeks to support teachers in improving classroom focus, enhancing the learning experience, and providing reliable performance evaluations that align with the goals of OBE, ensuring that
students meet targeted educational outcomes.

# PROPOSED SYSTEM

## Flowchart



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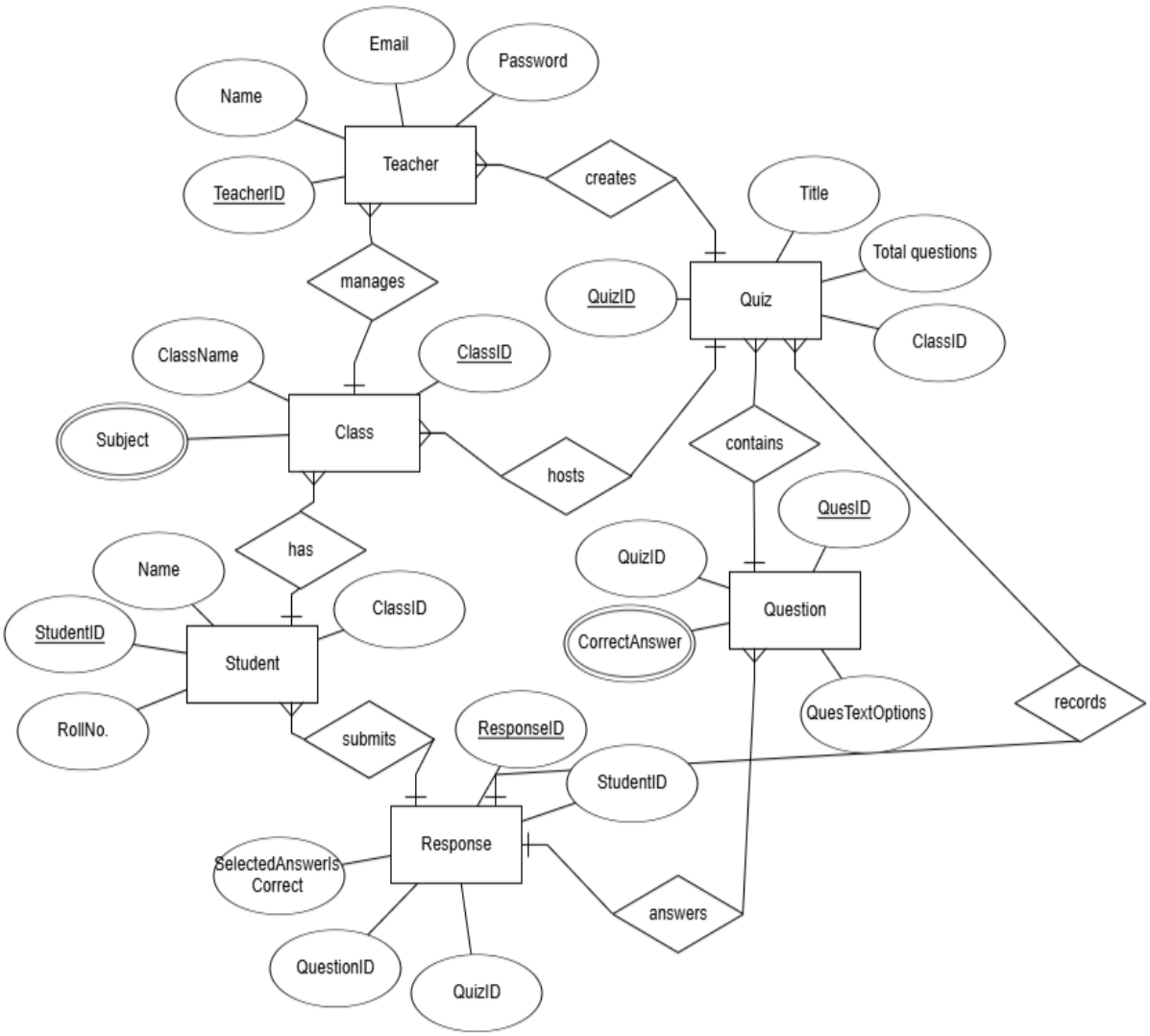
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Response
SelectedAnswerIs Correct
answers
QuestionID
QuizID

# Proposed Algorithm

1. **Start Process**
    - Initiate the RFID-based lecture assessment system.

2. **Teacher Delivers Lecture**
    - The teacher conducts the classroom session as usual.

3. **Provide Quiz**
    - After the lecture, the teacher announces a quiz session, assigning answer options (A/B/C/D) linked to specific RFID tags.

4. **Student Attempts Quiz using RFID Tags**
    - Each student chooses their answer by selecting the RFID tag corresponding to their selected option (e.g., RFID A for Option A).

5. **Submit Student Input via RFID Scanner**
    - Students scan their selected RFID tag at the centralized RFID scanner.
    - The system reads the tag and captures:
        - Student Unique ID
        - Selected Answer Option
        - Timestamp of submission

6. **Save to Database**
    - The system stores all response data securely in the database.

7. **Evaluate Responses**
    - The backend matches submitted responses with the correct answer key.
    - Marks are computed for each student accordingly.

8. **Fetch Data for Website/WebApp**
    - The frontend dashboard (WebApp) pulls the evaluated data from the database.

9. **Display Overall Performance**
    - The WebApp displays individual and collective performance reports.
    - Teachers and students can both view feedback.

10. **Teacher Plans Future Lectures**

- Based on quiz outcomes, the teacher modifies or reinforces specific lecture content in future sessions.

11. **End Process**

- The cycle concludes after storing results and planning follow-up actions.

# TECHNOLOGY USED

- **Frontend**
  **React.js:** A JavaScript library for building interactive user interfaces, providing dynamic and responsive web experiences for teachers and students.

- **Backend**
  **Node.js:** A runtime environment for building fast and scalable server-side applications, enabling real-time communication between the hardware device and the software.

- **Database**
  **MongoDB:** A NoSQL database that stores and manages student data, quiz results, and performance evaluations.

- **IoT**
  **RFID-IoT Integration:** Enables seamless communication between the RFID-based hardware device and the software system, facilitating real-time capture, transmission, and monitoring of student quiz responses during classroom assessments without the need for personal internet-enabled devices.

# EXPECTED OUTCOMES

1. **Improved Student Engagement**

   - Students actively participate in MCQ-based learning without distractions from mobile devices.
   - Increased attentiveness in class due to interactive learning.

2. **Enhanced Teacher Evaluation & Feedback**

   - Teachers can track individual student responses and analyse their performance in real-time.
   - Immediate feedback allows teachers to identify weak areas and adjust teaching methods accordingly.

3. **Mobile-Free Learning Environment**

   - Reduces dependency on smartphones during class, minimizing distractions.
   - Encourages focus on academic discussions rather than external interruptions.

4. **Increased Classroom Efficiency**

   - Saves time in conducting quizzes, tests, or polls.
   - Eliminates manual grading efforts with automated assessment.

5. **Data-Driven Decision Making**

   - The system collects and stores student response data for long-term analysis.
   - Helps in understanding learning patterns, improving future lessons.

# CONCLUSION

The development of this project aims to provide a solution to the growing challenge of mobile distractions in modern classrooms. By introducing a dedicated hardware device, students can participate in Multiple Choice Question (MCQ)-based assessments in a focused, controlled environment, free from the disruptions of mobile phones. This not only improves student engagement but also ensures that their responses are a more accurate reflection of their understanding, leading to better and more reliable evaluations by teachers.

The seamless integration of the hardware device with the software allows for real-time tracking of student performance. Teachers receive immediate feedback on each student's responses, enabling them to identify learning gaps quickly and adjust their teaching methods accordingly. This supports the principles of Outcome-Based Education (OBE) by allowing for continuous improvement and personalized learning, ensuring that educational objectives are met effectively.

As a result, this project enhances both the learning experience for students and the assessment process for teachers. By creating a distraction-free learning environment, it fosters more focused classroom activities, promotes active participation, and ensures a more accurate measurement of student progress. The outcome is a more efficient and outcome-driven educational setting, aligned with the goals of modern education.

# REFERENCES


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